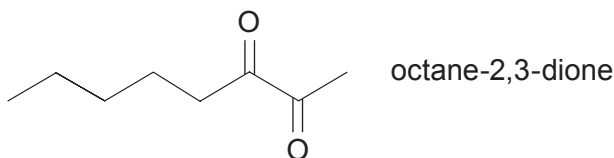


- (d) The mass spectrum of octane-2,3-dione (M_r 142) shows fragmentation peaks at m/z 99, 71 and 43.



Identify the species responsible for these peaks.

[2]

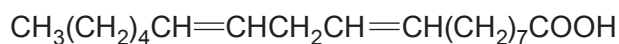
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- (e) Oct-1-en-3-ol is formed biochemically by the breakdown of linoleic acid.



linoleic acid

The carbon to carbon double bonds in linoleic acid are reduced by hydrogen and the resulting product can then be reduced by another suitable reducing agent to give a compound of formula $\text{C}_{18}\text{H}_{38}\text{O}$.

- (i) Write the **skeletal** formula of this compound.

[2]

- (ii) State the name of the reducing agent used in the second stage of the reduction.

[1]

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